

o/p:

```
{ _id: 1, name: 'xPhone', price: 799 },  
{ _id: 2, name: 'xTablet', price: 899 },  
{ _id: 3, name: 'SmartTablet', price: 899 }  
]
```

```
db.products.find(  
  "spec.screen": {  
    $gte: 9.5  
  }  
, {  
  name: 1,  
  "spec.screen": 1  
)
```

o/p:

```
[  
  { _id: 2, name: 'xTablet', spec: { screen: 9.5 } },  
  { _id: 3, name: 'SmartTablet', spec: { screen: 9.7 } },  
  { _id: 4, name: 'SmartPad', spec: { screen: 9.7 } },  
  { _id: 5, name: 'SmartPhone', spec: { screen: 9.7 } }  
]
```

```
db.products.find(  
  storage: {  
    $gte: 512  
  }  
, {  
  name: 1,  
  storage: 1  
)
```

o/p:

```
[  
  { _id: 2, name: 'xTablet', storage: [ 128, 256, 512 ] },  
  { _id: 4, name: 'SmartPad', storage: [ 128, 256, 1024 ] }  
]
```

```
db.products.find(  
  "releaseDate": {  
    $gte: new ISODate('2020-05-14')
```

```

    }
  }, {
    name: 1,
    releaseDate: 1
  });

```

```

o/p: [
  {
    _id: 4,
    name: 'SmartPad',
    releaseDate: ISODate("2020-05-14T00:00:00.000Z")
  },
  {
    _id: 5,
    name: 'SmartPhone',
    releaseDate: ISODate("2022-09-14T00:00:00.000Z")
  }
]

```

```

db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate":
ISODate("2011-05-14"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
2.66 }, "color":["white","black"],"storage":[64,128,256]},
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate":
ISODate("2011-09-01") , "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
3.66 }, "color":["white","black","purple"],"storage":[128,256,512]},
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate":
ISODate("2015-01-14"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
3.66 }, "color":["blue"],"storage":[16,64,128]},
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate":
ISODate("2020-05-14"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256,1024]},
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate":
ISODate("2022-09-14"), "spec" : { "ram" : 4, "screen" : 9.7, "cpu" :
1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256]}
,
  { "_id" : 6, "name" : "xWidget", "spec" : { "ram" : 64, "screen" : 9.7, "cpu" :
3.66 }, "color":["black"],"storage":[1024]}
])

```

```

db.products.find({
  price: {
    $ne: 899
  }
}, {
  name: 1,

```

```
    price: 1
  })
```

```
o/p:[
  { _id: 1, name: 'xPhone', price: 799 },
  { _id: 4, name: 'SmartPad', price: 699 },
  { _id: 5, name: 'SmartPhone', price: 599 },
  { _id: 6, name: 'xWidget' }
]
```

```
db.products.find({
  "spec.screen": {
    $ne: 9.7
  }
}, {
  name: 1,
  "spec.screen": 1
})
```

```
[
  { _id: 1, name: 'xPhone', spec: { screen: 6.5 } },
  { _id: 2, name: 'xTablet', spec: { screen: 9.5 } }
]
```

```
db.products.find({
  storage: {
    $ne: 128
  }
}, {
  name: 1,
  storage: 1
});
```

```
o/p:[ { _id: 6, name: 'xWidget', storage: [ 1024 ] } ]
```

```
db.products.find({
  releaseDate: {
    $ne: new ISODate('2015-01-14')
  }
}, {
  name: 1,
  releaseDate: 1
});
```

```
o/p:[
```

```

{
  _id: 1,
  name: 'xPhone',
  releaseDate: ISODate("2011-05-14T00:00:00.000Z")
},
{
  _id: 2,
  name: 'xTablet',
  releaseDate: ISODate("2011-09-01T00:00:00.000Z")
},
{
  _id: 4,
  name: 'SmartPad',
  releaseDate: ISODate("2020-05-14T00:00:00.000Z")
},
{
  _id: 5,
  name: 'SmartPhone',
  releaseDate: ISODate("2022-09-14T00:00:00.000Z")
},
{ _id: 6, name: 'xWidget' }
]

```

```

db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate":
ISODate("2011-05-14"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
2.66 }, "color":["white","black"],"storage":[64,128,256]},
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate":
ISODate("2011-09-01") , "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
3.66 }, "color":["white","black","purple"],"storage":[128,256,512]},
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate":
ISODate("2015-01-14"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
3.66 }, "color":["blue"],"storage":[16,64,128]},
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate":
ISODate("2020-05-14"),"spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256,1024]},
  { "_id" : 5, "name" : "SmartPhone", "price" : 599,"releaseDate":
ISODate("2022-09-14"), "spec" : { "ram" : 4, "screen" : 5.7, "cpu" :
1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256]}
])

```

```

db.products.find({
  price: {
    $in: [699, 799]
  }
}, {

```

```
    name: 1,  
    price: 1  
  })
```

o/p:

```
[  
  { _id: 1, name: 'xPhone', price: 799 },  
  { _id: 4, name: 'SmartPad', price: 699 }  
]
```

```
db.products.find({  
  color: {  
    $in: ["black", "white"]  
  }  
}, {  
  name: 1,  
  color: 1  
})
```

o/p:

```
[  
  { _id: 1, name: 'xPhone', color: [ 'white', 'black' ] },  
  { _id: 2, name: 'xTablet', color: [ 'white', 'black', 'purple' ] },  
  {  
    _id: 4,  
    name: 'SmartPad',  
    color: [ 'white', 'orange', 'gold', 'gray' ]  
  },  
  {  
    _id: 5,  
    name: 'SmartPhone',  
    color: [ 'white', 'orange', 'gold', 'gray' ]  
  }  
]
```

```
db.products.find({  
  color: {  
    $in: [/^g+/, /^w+/]  
  }  
}, {  
  name: 1,  
  color: 1  
})
```

o/p:

```
[
  { _id: 1, name: 'xPhone', color: [ 'white', 'black' ] },
  { _id: 2, name: 'xTablet', color: [ 'white', 'black', 'purple' ] },
  {
    _id: 4,
    name: 'SmartPad',
    color: [ 'white', 'orange', 'gold', 'gray' ]
  },
  {
    _id: 5,
    name: 'SmartPhone',
    color: [ 'white', 'orange', 'gold', 'gray' ]
  }
]
```

```
db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate":
  ISODate("2011-05-14"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
  2.66 }, "color":["white","black"],"storage":[64,128,256]},
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate":
  ISODate("2011-09-01") , "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
  3.66 }, "color":["white","black","purple"],"storage":[128,256,512]},
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate":
  ISODate("2015-01-14"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
  3.66 }, "color":["blue"],"storage":[16,64,128]},
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate":
  ISODate("2020-05-14"),"spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
  1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256,1024]},
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate":
  ISODate("2022-09-14"), "spec" : { "ram" : 4, "screen" : 5.7, "cpu" :
  1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256]}
])
```

```
db.products.find({
  price: {
    $nin: [699, 799]
  }
}, {
  name: 1,
  price: 1
})
```

o/p:[
{ _id: 2, name: 'xTablet', price: 899 },

```
{ _id: 3, name: 'SmartTablet', price: 899 },
{ _id: 5, name: 'SmartPhone', price: 599 }
]
```

```
db.products.find({
  color: {
    $nin: ["black", "white"]
  }
}, {
  name: 1,
  color: 1
})
```

```
o/p:[ { _id: 3, name: 'SmartTablet', color: [ 'blue' ] } ]
```

```
db.products.find({
  color: {
    $nin: [/^g+/, /^w+/]
  }
}, {
  name: 1,
  color: 1
})
```

```
o/p:[ { _id: 3, name: 'SmartTablet', color: [ 'blue' ] } ]
```

```
db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate" :
ISODate("2011-05-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
2.66 }, "color" : [ "white", "black" ], "storage" : [ 64, 128, 256 ] },
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate" :
ISODate("2011-09-01T00:00:00Z"), "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
3.66 }, "color" : [ "white", "black", "purple" ], "storage" : [ 128, 256, 512 ] },
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate" :
ISODate("2015-01-14T00:00:00Z"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
3.66 }, "color" : [ "blue" ], "storage" : [ 16, 64, 128 ] },
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate" :
ISODate("2020-05-14T00:00:00Z"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256,
1024 ] },
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate" :
ISODate("2022-09-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 9.7, "cpu" :
1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256 ] },
  { "_id" : 6, "name" : "xWidget", "spec" : { "ram" : 64, "screen" : 9.7, "cpu" :
3.66 }, "color" : [ "black" ], "storage" : [ 1024 ] }
])
```

```

db.products.find({
  $and: [{
    price: 899
  }, {
    color: {
      $in: ["white", "black"]
    }
  }]
}, {
  name: 1,
  price: 1,
  color: 1
})

```

```

o/p: [
  {
    _id: 2,
    name: 'xTablet',
    price: 899,
    color: [ 'white', 'black', 'purple' ]
  }
]

```

```

db.products.find({
  $and: [{
    price: 699
  }, {
    price: {
      $exists: true
    }
  }]
}, {
  name: 1,
  price: 1,
  color: 1
})

```

```

o/p: [
  {
    _id: 4,
    name: 'SmartPad',
    price: 699,
    color: [ 'white', 'orange', 'gold', 'gray' ]
  }
]

```


]

```
db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate" :
    ISODate("2011-05-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
    2.66 }, "color" : [ "white", "black" ], "storage" : [ 64, 128, 256 ] },
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate" :
    ISODate("2011-09-01T00:00:00Z"), "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
    3.66 }, "color" : [ "white", "black", "purple" ], "storage" : [ 128, 256, 512 ] },
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate" :
    ISODate("2015-01-14T00:00:00Z"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
    3.66 }, "color" : [ "blue" ], "storage" : [ 16, 64, 128 ] },
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate" :
    ISODate("2020-05-14T00:00:00Z"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
    1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256,
    1024 ] },
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate" :
    ISODate("2022-09-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 9.7, "cpu" :
    1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256 ] }
])
```

```
db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate":
    ISODate("2011-05-14"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
    2.66 }, "color":["white","black"],"storage":[64,128,256]},
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate":
    ISODate("2011-09-01") , "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
    3.66 }, "color":["white","black","purple"],"storage":[128,256,512]},
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate":
    ISODate("2015-01-14"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
    3.66 }, "color":["blue"],"storage":[16,64,128]},
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate":
    ISODate("2020-05-14"),"spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
    1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256,1024]},
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate":
    ISODate("2022-09-14"), "spec" : { "ram" : 4, "screen" : 5.7, "cpu" :
    1.66 }, "color":["white","orange","gold","gray"],"storage":[128,256]}
])
```

```
db.products.find({
  price: {
    $gte: 799
  }
}, {
  name: 1,
  price: 1
})
```

D

```
db.products.find({
  $or: [{
    price: 799
  }, {
    price: 899
  }]
}, {
  name: 1,
  price: 1
})
```

```
o/p: [
  { _id: 1, name: 'xPhone', price: 799 },
  { _id: 2, name: 'xTablet', price: 899 },
  { _id: 3, name: 'SmartTablet', price: 899 }
]
```

```
db.products.find({
  price: {
    $in: [799, 899]
  }
}, {
  name: 1,
  price: 1
})
```

```
db.products.find({
  $or: [
    { price: { $lt: 699 } },
    { price: { $gt: 799 } }
  ]
}, {
  name: 1,
  price: 1
})
```

```
o/p: [
  { _id: 2, name: 'xTablet', price: 899 },
  { _id: 3, name: 'SmartTablet', price: 899 },
  { _id: 5, name: 'SmartPhone', price: 599 }
]
```

```

db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate" :
    ISODate("2011-05-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
    2.66 }, "color" : [ "white", "black" ], "storage" : [ 64, 128, 256 ] },
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate" :
    ISODate("2011-09-01T00:00:00Z"), "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
    3.66 }, "color" : [ "white", "black", "purple" ], "storage" : [ 128, 256, 512 ] },
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate" :
    ISODate("2015-01-14T00:00:00Z"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
    3.66 }, "color" : [ "blue" ], "storage" : [ 16, 64, 128 ] },
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate" :
    ISODate("2020-05-14T00:00:00Z"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
    1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256,
    1024 ] },
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate" :
    ISODate("2022-09-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 9.7, "cpu" :
    1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256 ] },
  { "_id" : 6, "name" : "xWidget", "spec" : { "ram" : 64, "screen" : 9.7, "cpu" :
    3.66 }, "color" : [ "black" ], "storage" : [ 1024 ] }
])

```

```

db.products.find({
  price: {
    $not: {
      $gt: 699
    }
  }
}, {
  name: 1,
  price: 1
})

```

```

o/p: [
  { _id: 4, name: 'SmartPad', price: 699 },
  { _id: 5, name: 'SmartPhone', price: 599 },
  { _id: 6, name: 'xWidget' }
]

```

```

db.products.find({
  price: {
    $lte: 699
  }
}, {
  name: 1,
  price: 1
})

```

```
o/p: [
  { _id: 4, name: 'SmartPad', price: 699 },
  { _id: 5, name: 'SmartPhone', price: 599 }
]
```

```
db.products.find({
  name: {
    $not: /^Smart+/
  }
}, {
  name: 1
})
```

```
o/p: [
  { _id: 1, name: 'xPhone' },
  { _id: 2, name: 'xTablet' },
  { _id: 6, name: 'xWidget' }
]
```

```
db.products.insertMany([
  { "_id" : 1, "name" : "xPhone", "price" : 799, "releaseDate" :
  ISODate("2011-05-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 6.5, "cpu" :
  2.66 }, "color" : [ "white", "black" ], "storage" : [ 64, 128, 256 ] },
  { "_id" : 2, "name" : "xTablet", "price" : 899, "releaseDate" :
  ISODate("2011-09-01T00:00:00Z"), "spec" : { "ram" : 16, "screen" : 9.5, "cpu" :
  3.66 }, "color" : [ "white", "black", "purple" ], "storage" : [ 128, 256, 512 ] },
  { "_id" : 3, "name" : "SmartTablet", "price" : 899, "releaseDate" :
  ISODate("2015-01-14T00:00:00Z"), "spec" : { "ram" : 12, "screen" : 9.7, "cpu" :
  3.66 }, "color" : [ "blue" ], "storage" : [ 16, 64, 128 ] },
  { "_id" : 4, "name" : "SmartPad", "price" : 699, "releaseDate" :
  ISODate("2020-05-14T00:00:00Z"), "spec" : { "ram" : 8, "screen" : 9.7, "cpu" :
  1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256,
  1024 ] },
  { "_id" : 5, "name" : "SmartPhone", "price" : 599, "releaseDate" :
  ISODate("2022-09-14T00:00:00Z"), "spec" : { "ram" : 4, "screen" : 9.7, "cpu" :
  1.66 }, "color" : [ "white", "orange", "gold", "gray" ], "storage" : [ 128, 256 ] },
  { "_id" : 6, "name" : "xWidget", "spec" : { "ram" : 64, "screen" : 9.7, "cpu" :
  3.66 }, "color" : [ "black" ], "storage" : [ 1024 ] }
])
```

```
db.products.find({
  $nor:[
    { price: 899},
    { color: "gold"}
  ]
}, {
```

```
    name: 1,  
    price: 1,  
    color: 1  
  })
```

```
{ "_id" : 1, "name" : "xPhone", "price" : 799, "color" : [ "white", "black" ] }  
{ "_id" : 6, "name" : "xWidget", "color" : [ "black" ] }
```